5.3.2 Definite & Indefinite Integrals

Motivation: In the last section we saw that antiderivatives are important for computing definite integrals. We would like some formal notation for the most general antiderivative of a function.

Definition: We use an **indefinite integral** of *f*, written as $\int f(x)\,dx$, to indicate the most general antiderivative of *f*.

Example 1: Find each of the following.

a) Find $\int x^2 dx$.

b) Find $\int_0^3 x^2 dx$

Note: VERY IMPORTANTLY, we see that an indefinite integral represents a family of functions, while a definite integral represents an area

Properties of Indefinite Integrals (Antiderivatives)

(1) $\int c \cdot f(x)\,dx = c \cdot \int f(x)\,dx$ (constant multiple rule in reverse)

(2) $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$ (sum & difference rule in reverse)

(3) $\int f(x)g(x)\,dx \neq \int f(x)\,dx \cdot \int g(x)\,dx$ (you *cannot* take the antiderivative of each piece of a product) This is *NOT* a derivative rule in reverse.

In the online textbook there is a link to what is called the "Table of Integrals" and it is very important to know ALL of these formulas for Calculus I, II, & III. All of these formulas simply represent "differentiation in reverse".

General Strategies for Finding Antiderivatives

(1) Use a derivative rule in reverse.

(2) Use Algebra to simplify your expression and then use (1).

Note: We will see here that "Algebra" includes things such as simplifying expressions, factoring, using trig. identities, and a variety of other simplification techniques.

Note: The list above is by no means complete. We will continue to add to this list throughout the remainder of Calculus I and during Calculus II.

Example 2: Evaluate $\int \frac{1}{\sqrt{1-x^2}}\,dx$.

Note: In Examples 3 – 6 (below) a variety of different "Algebra" techniques are shown.

Example 3: Evaluate $\int \frac{x^3 - 16x^4 \sec^2(x) + 8}{8x^4}\,dx$. ***(Split Fractions)***

Example 4: Evaluate $\int_1^2 \frac{x^2+4x+3}{x+1}\,dx$. ***(Factoring)***

Example 5: Evaluate $\int \frac{x^2+10}{x^2+1}\,dx$. ***(Split fractions cleverly)***

Example 6: Evaluate $\int_{-1}^0 \frac{x^2}{x^2+1}\,dx$. ***(Introduce a "fancy 0")***

Important Trig. Identities to Know:

- Pythagorean Identities
    - $\sin^2(x)+\cos^2(x)=1$
    - $\tan^2(x)+1=\sec^2(x)$
    - $1+\cot^2(x)=\csc^2(x)$

- Double/Half Angle Identities
    - $\sin(2x)=2\sin(x)\cos(x)$
    - $\cos(2x)=\cos^2(x)-\sin^2(x)$

Example 7: Evaluate $\int \sec(x)\left[\sec(x)+\tan(x)\right]dx$.

Example 8: Evaluate $\int \tan^2(x)\,dx$.

Example 9: Evaluate $\displaystyle\int_{\pi/4}^{\pi/2} \frac{\cos(\theta)}{\sin^2(\theta)}\,d\theta$.